

Community-level Risk Mapping and Vulnerability Assessments (VRAM) Presenting to WHO - Team

By

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Presentation Outlines

- 1. Key considerations
- 2. Main activities
- 3. Potential challenges.

- I. Key Consideration
 - Population distribution and Population density
 - U5, 5-14, 15-45, 45+
 - The concentration of population vs. chemical industry vs. IDSR data systematically reported using
 - the political administrative level of region(provinces), zones, districts and villages, health facilities (based on the population catchment area),
 - Data of IDSR (Surveillance): at least a retrospective data 5-10 years (Systematically collected based on a weekly based)
 - Data of GIS Shape files
 - Location of Health facilities, Hospitals, health posts, rivers, roads, altitude, water point,
 - Softwares ARCGIS and Statistical packages softwares
 - GPS to collect some missing GIS files including point data, road data, river data and some water points,
 - Political administrative levels data flow from HFs, district, Zone, regions (Province),
 - Location of Chemical industry Sites
 - CORIN analysis whether the land is industry zone, cultivation zone, urban zone or residential , buildings and etc ...
 - Location of all facilities including HFs, schools, hotels, gas station the chemical zones to identify risk,

II. Main activities

- Using the VRAM techniques producing of risk mapping for that specific community using
 - Surveillance data could be for example
 - Malaria data
 - AWD/Cholera data and etc ...
 - Focusing on the major disease burden in the community or high risk place like chemical industry place including for example
 - Refugee camp main disease burden this is a specific place
 - General disease burden like : Ebola, Meningitis, Malaria, flood prone and etc...
 - Based on the above Surveillance data, we can look in depth the map correlation and association of that specific geographic place with the reported diseases or the location of the chemical industry or the vs. the altitude, rain precipitation, the location of water body and etc ... (Generally what is the association/ correlation specific event happening in one geographic place vs. with a number of IDSR data (the (association),
 - Using different layers calculating **risk level by different kilometer radius**
 - **One layer** could be population density by different political administrative level
 - **Another layer** the location of HFs (Shows equity)
 - **3rd layer** could be the road vs. the above two (shows accessibility),
 - And etc ...
 - Water coverage and latrine coverage by different political administrative level
Water coverage by regions, zones, districts villages. It has a risk of water borne diseases, water related diseases, water based diseases and water scarce diseases (respectively like AWD, typhoid, malaria, yellow fever, river blindness, sleeping sickness, guinea worm, bilharzia, trachoma and scabies),

III. Potential Challenges

- Sometimes, some countries have poor **data repository** sometimes back to 5-10 years
- Some GIS data missing and cost a lot of time, energy, cost to collect (Most of the GIS data freely available in the HDX humanitarian web),
- Some places missing important equipment and softwares including :
 - Computers
 - Plotter machines
 - ARCGIS Softwares
- Human resource gap
 - Most of the time no trained individual on GIS and Mapping. Even capacity building training requires some background,
- **Ownership issues** at the MoH in a given country, **capacity building can be given** but they have to be ready to accept the VRAM process and **they have to cooperate to make ready some data**, (Data belongs to government in the respective country MoH), based on IHR (Data Sharing policy),

IV. Recommendations

- GIS Shape files needs to be ready from X country including :
 - Point data f HFs,
 - Road vs. HFs location (**accessibility**),
 - HFs vs. Population density (**equity**),
- Surveillance Data
 - **Communicable diseases** reported data using Weekly Epidemiological week
 - **Daily IDSR** (immediately reportable diseases,
 - **Non-communicable** reportable diseases n along the road), and accessibility (Could be weekly ...)
 - Location of Water body,
 - Water coverage and laterite coverage,
 - **water-borne diseases** (e.g. AWD, typhoid)
 - **water –related diseases** (e.g. malaria, yellow fever, river blindness, sleeping sickness)
 - **water-based diseases** (e.g. guinea worm and bilharzia) and water-scarce diseases (trachoma and scabies),
 - The final risk mapping shows **resource availability (gaps),** to **appropriate decision making** based on informed decision making, to **put appropriate resource allocation,** **to put appropriate control measures,**

Sample Works

- **From Four (#4) WHO Regions:**

- **AFRO**

- HIV Prevalence from 1992-2012 (20 yrs.)
- Ethiopia (WHO-AFRO)
- Sierra Leone (WHO – AFRO)

- **EMRO**

- Jordan (WHO-EMRO)

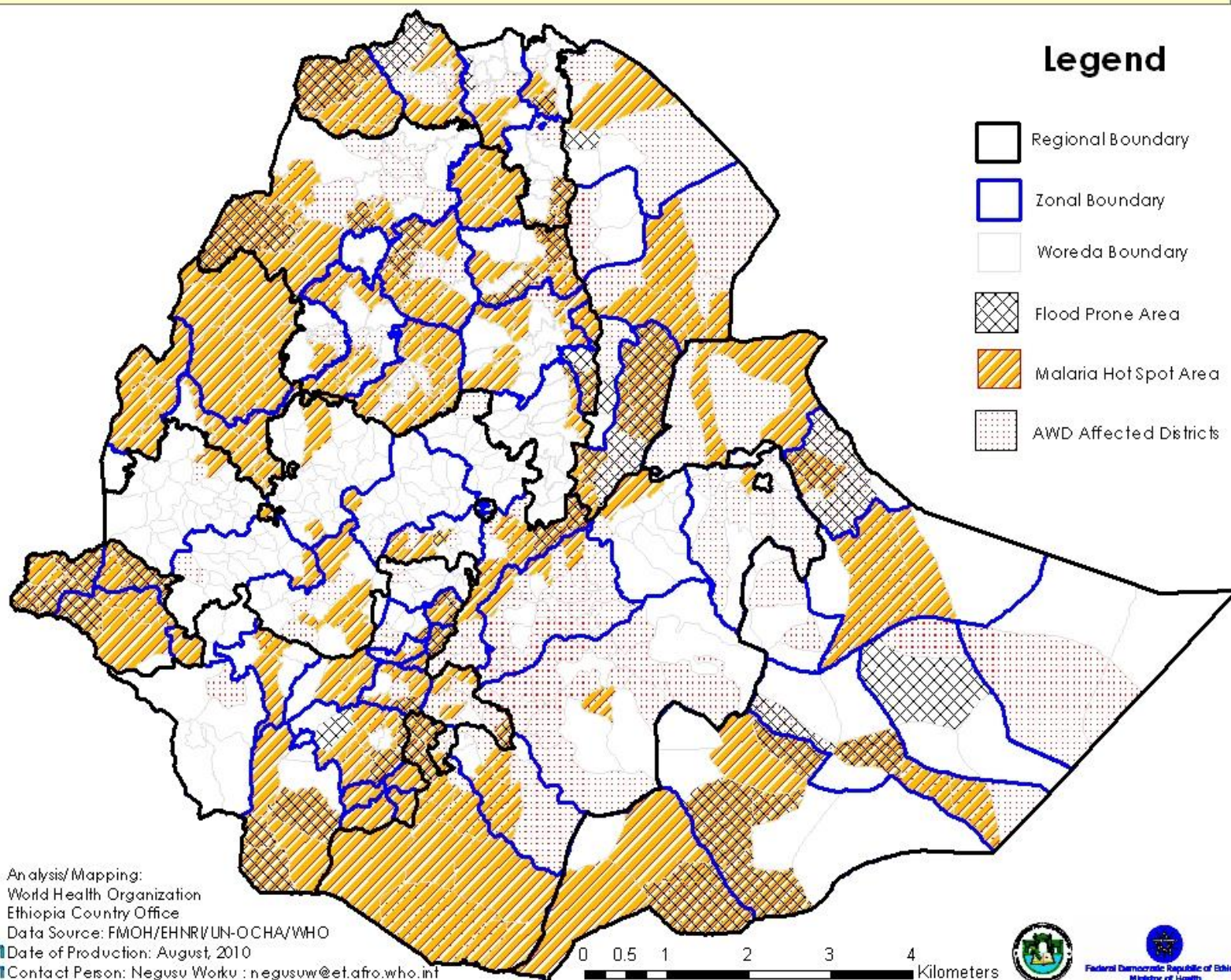
- **SEARO**

- Bangladesh (WHO-SEARO)
- India
- Afghanistan

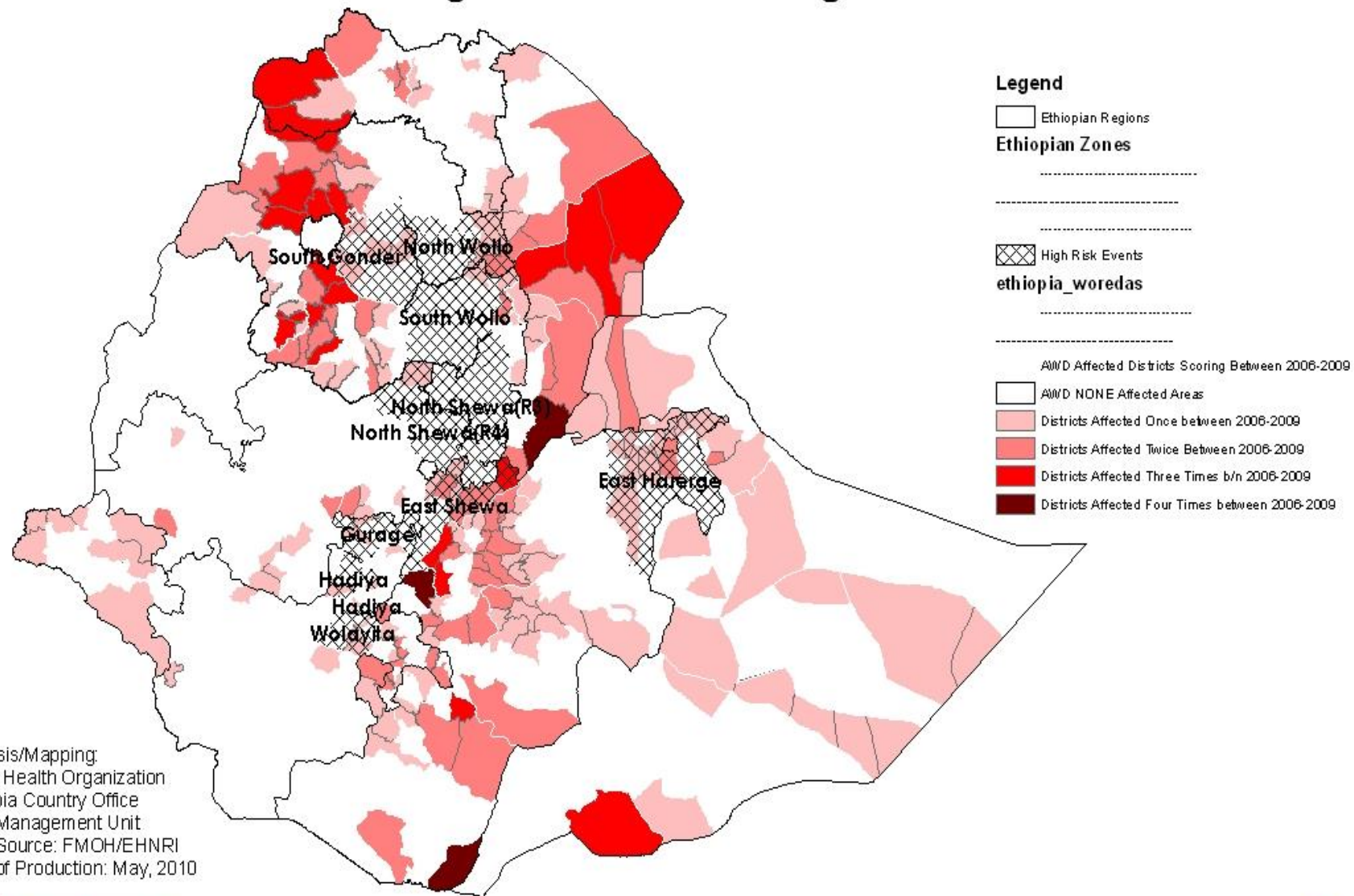
- **EURO**

- Turkey (WHO-EURO)

Map of Ethiopia Showing Correlation of AWD, Malaria and Flood Prone area as of August, 2010



Map of Ethiopia Showing AWD Affected Districts from 2006-2009 along With the National High Risk Events



Analysis/Mapping:
World Health Organization
Ethiopia Country Office
Data Management Unit
Data Source: FMOH/EHNRI
Date of Production: May, 2010

**Reported Malaria Caseload by PHUs
from Epidemic Week 43 - 51 (October 19 - December 20, 2015),
Port Loko, Sierra Leone**

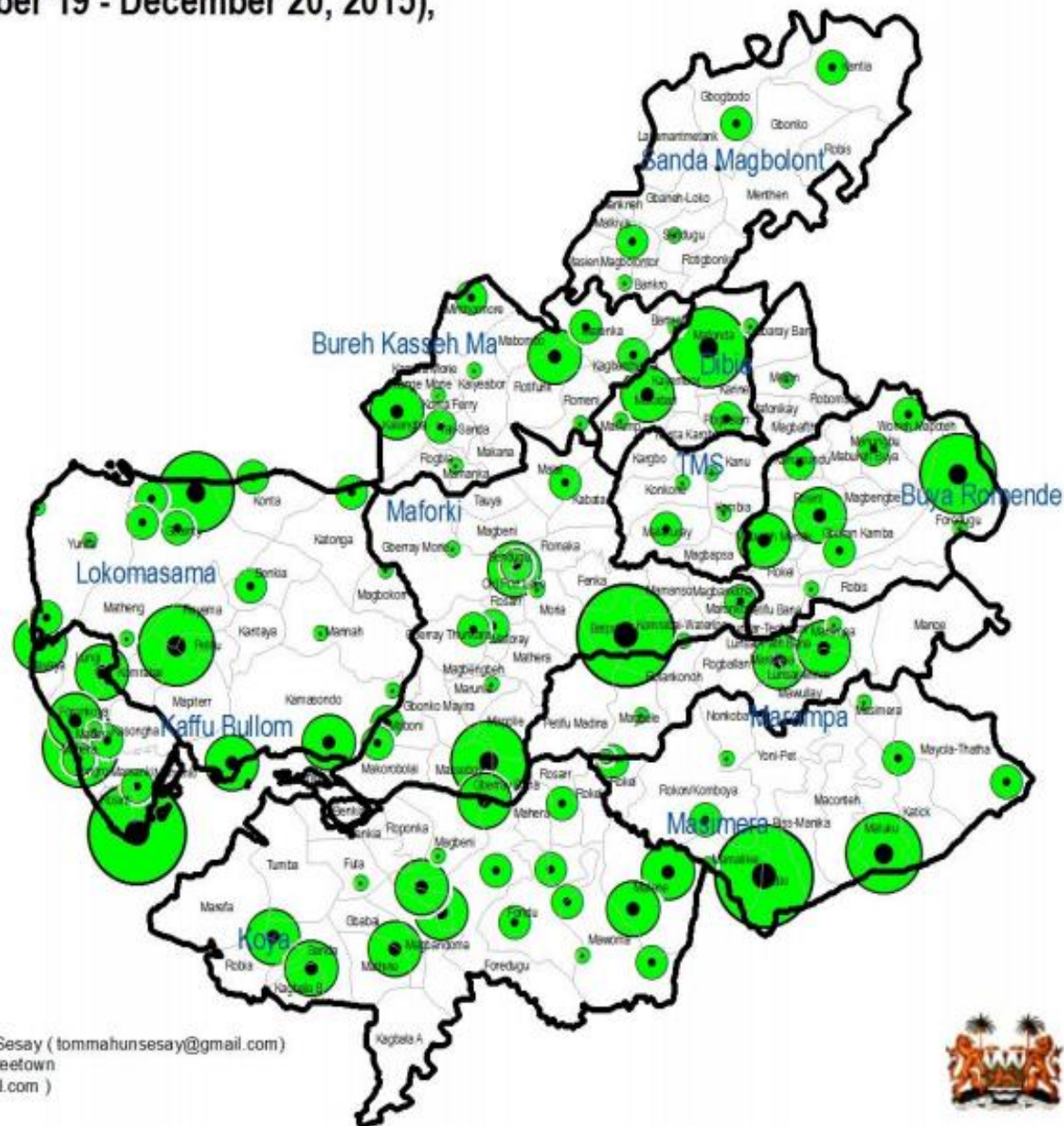
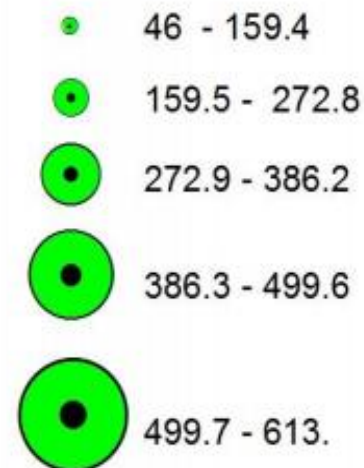
Legend



Port Loko Chiefdoms

PLK Sections

Malaria Caseload



Data Source : DHMT Port Loko
Contact : DMO - Port Loko : Dr. Tom Sesay (tommahunsesay@gmail.com)
Data Analysis and Mapping : WHO-Freetown
Negusu Worku (negusworku@gmail.com)
Date of Production: December, 2015

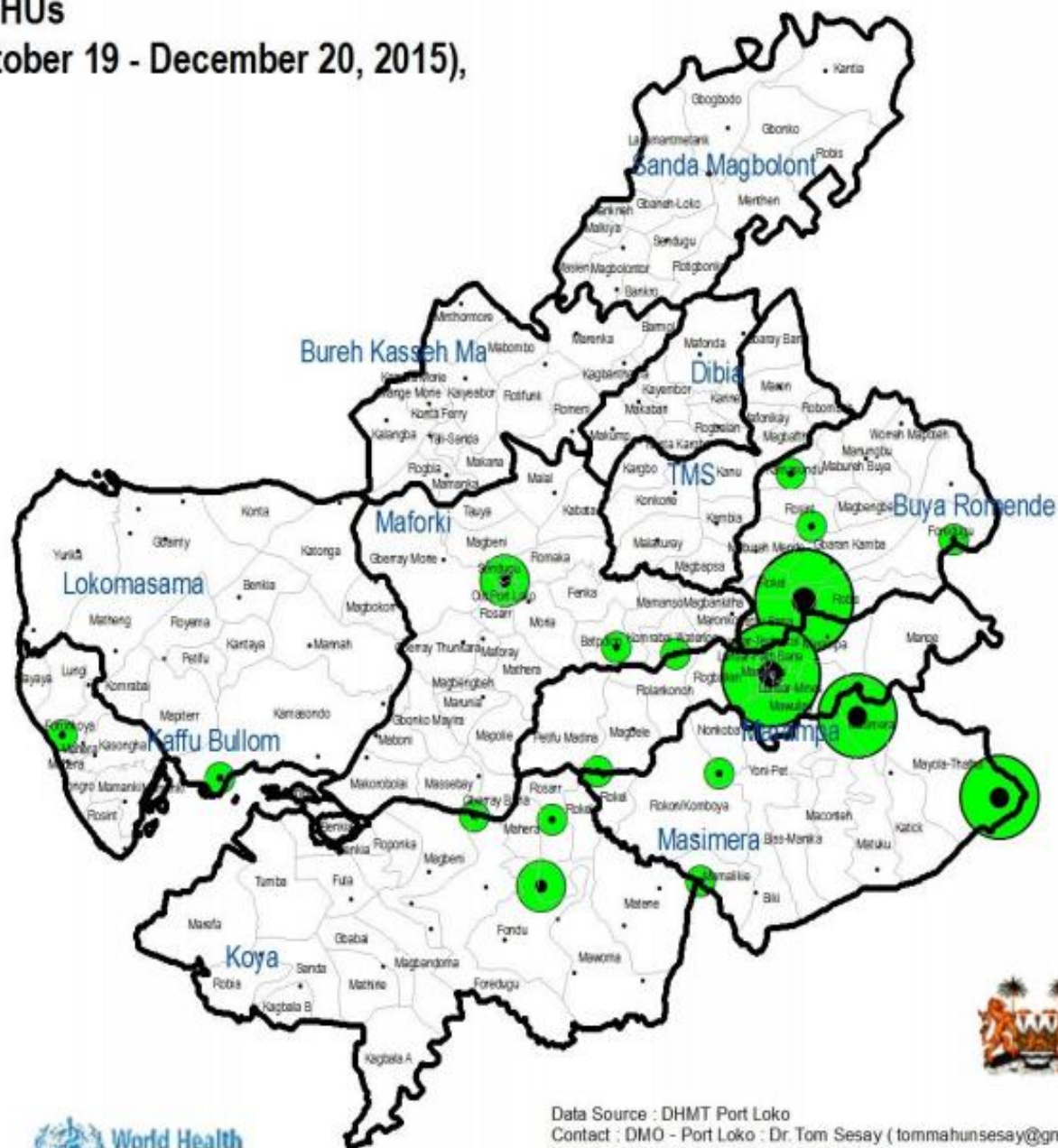
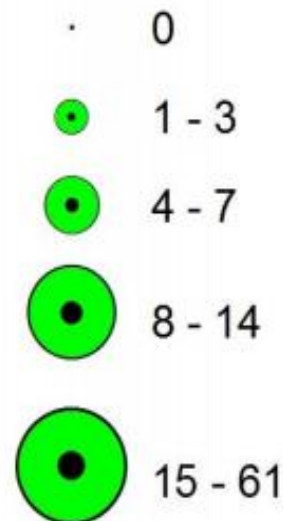


Reported Measles Caseload by PHUs from Epidemic Week 43 - 51 (October 19 - December 20, 2015), Port Loko, Sierra Leone

Legend

-  Port Loko Chiefdoms
-  PLK_Sections

Port Loko Measles



Reported Diarrhoea Severe Dehydration Caseload by PHUs from Epidemic Week 43 - 51 (October 19 - December 20, 2015), Port Loko, Sierra Leone

Legend

 Port Loko Chiefdoms

 PLK_Sections

-

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1 - 10



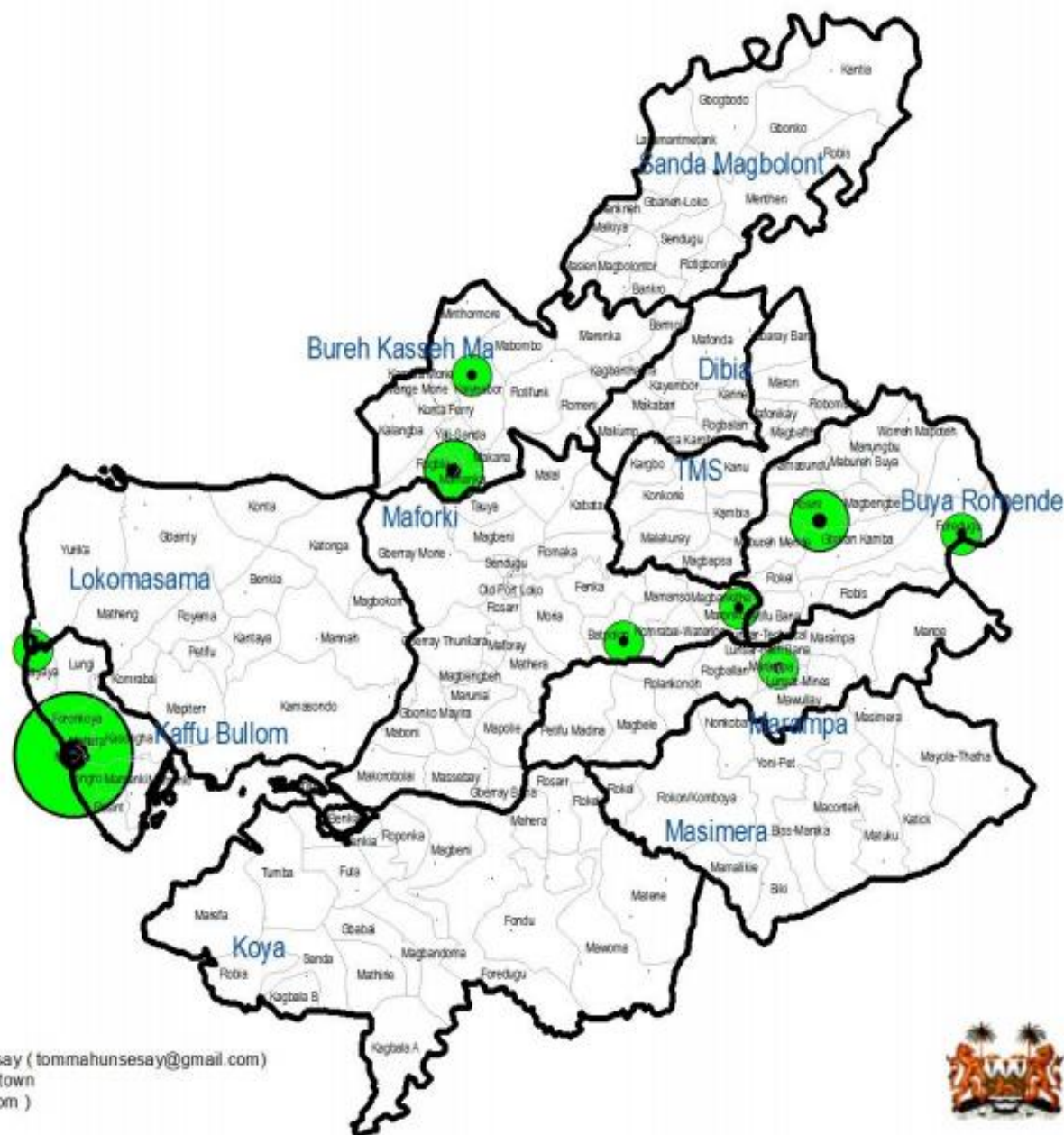
11 - 20



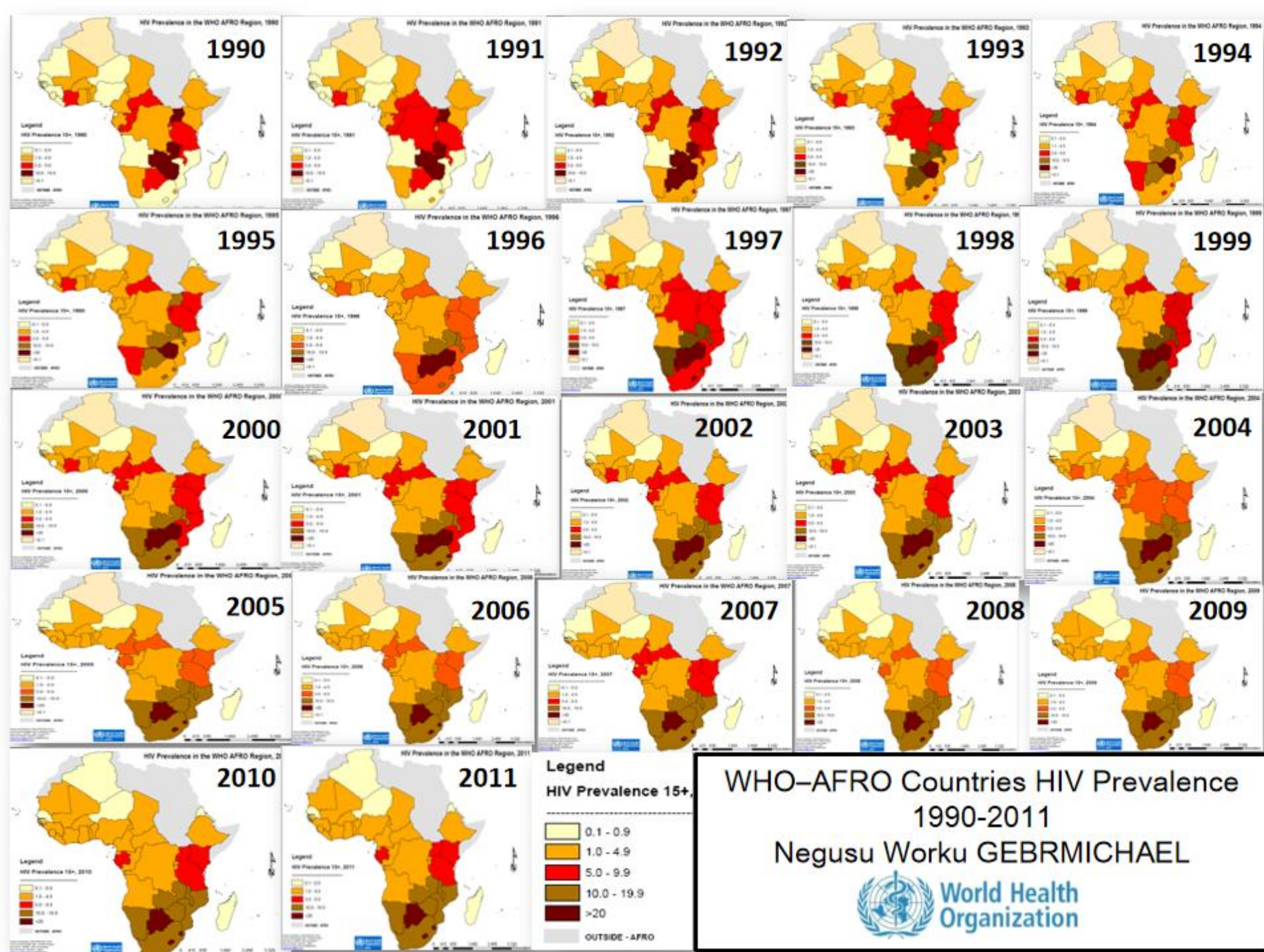
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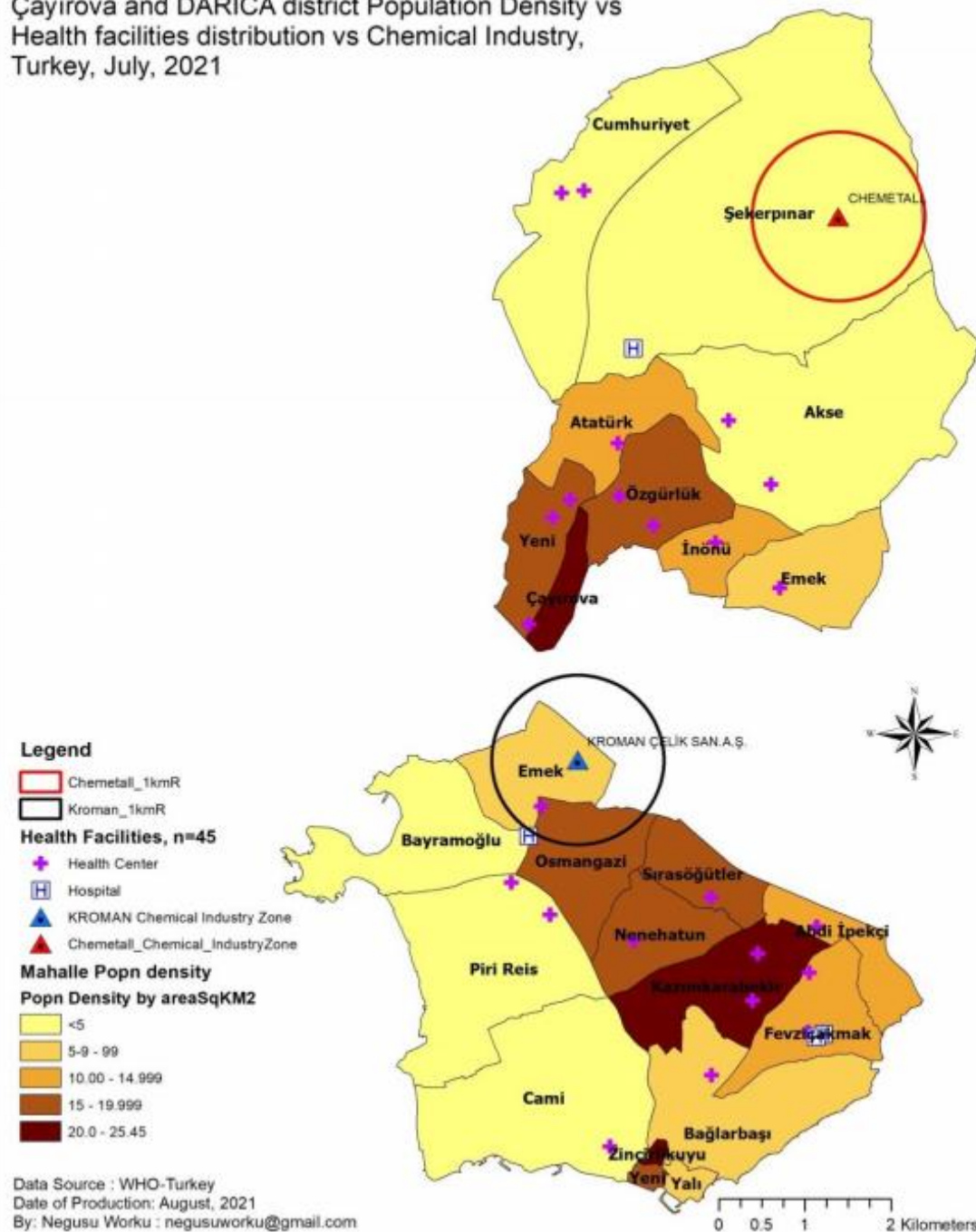
31 - 36



Data Source : DHMT Port Loko
Contact : DMO - Port Loko : Dr. Tom Sesay (tomahunesay@gmail.com)
Data Analysis and Mapping : WHO-Freetown
Negusu Worku (negusuworku@gmail.com)
Date of Production : December, 2015



Çayırova and DARICA district Population Density vs Health facilities distribution vs Chemical Industry, Turkey, July, 2021



Health Facilities (Health Center, Hospital and Pharmacy) shows n=44 Observations :

1. Health Center, n=25
2. Hospitals, n=4
3. Pharmacy, n=15

1. Çayırova district : health facilities availability 39% (n=1 hospital, n=12 Health Center and n=3 Hospitals),

2. Darica district : health facilities availability 61% (n=3 Hospitals, n=13 health centers, and n=11 Pharmacy),

3. Çayırova district, within 1kmR no facilities located,

4. Darica, within 1 kmR three (3) health centers available,

5. Çayırova district , within 2kmR radius no Health facilities are available,

6. Darica, within 2kmR, n=10 health facilities are available, of which n=3 Health Centers, n=1 hospitals and n=6 Pharmacies are available.

Çayırova and DARICA district Population Density vs Health facilities distribution vs Chemical Industry, by different risk km radius Turkey, July, 2021

Legend

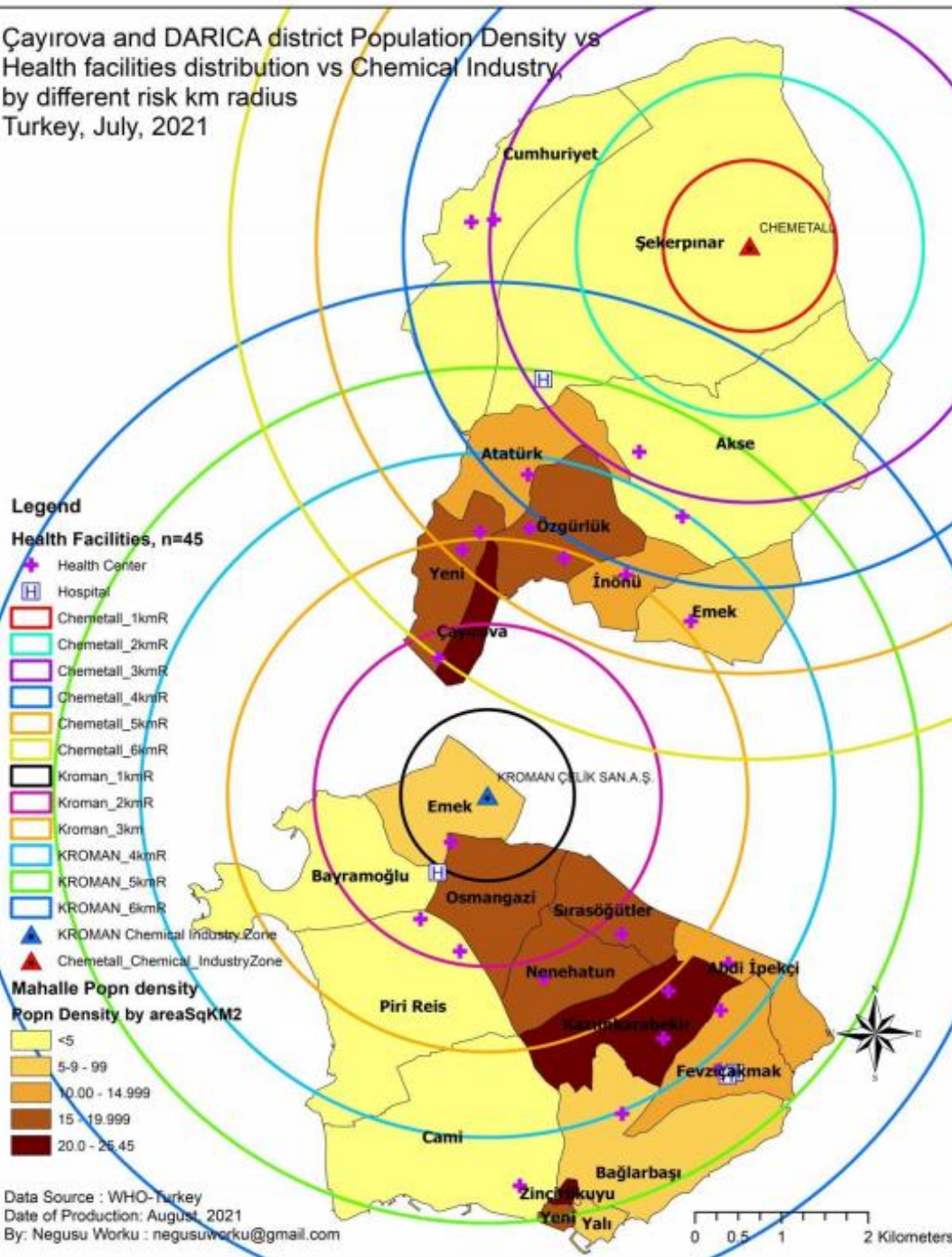
Health Facilities, n=45

- Health Center
- Hospital
- Chemetall_1kmR
- Chemetall_2kmR
- Chemetall_3kmR
- Chemetall_4kmR
- Chemetall_5kmR
- Chemetall_6kmR
- Kroman_1kmR
- Kroman_2kmR
- Kroman_3km
- KROMAN_4kmR
- KROMAN_5kmR
- KROMAN_6kmR
- KROMAN Chemical Industry Zone
- Chemetall_Chemical IndustryZone

Mahalle Popn density Popn Density by areaSqKM2

- <5
- 5-9 - 99
- 10.00 - 14.999
- 15 - 19.999
- 20.0 - 25.45

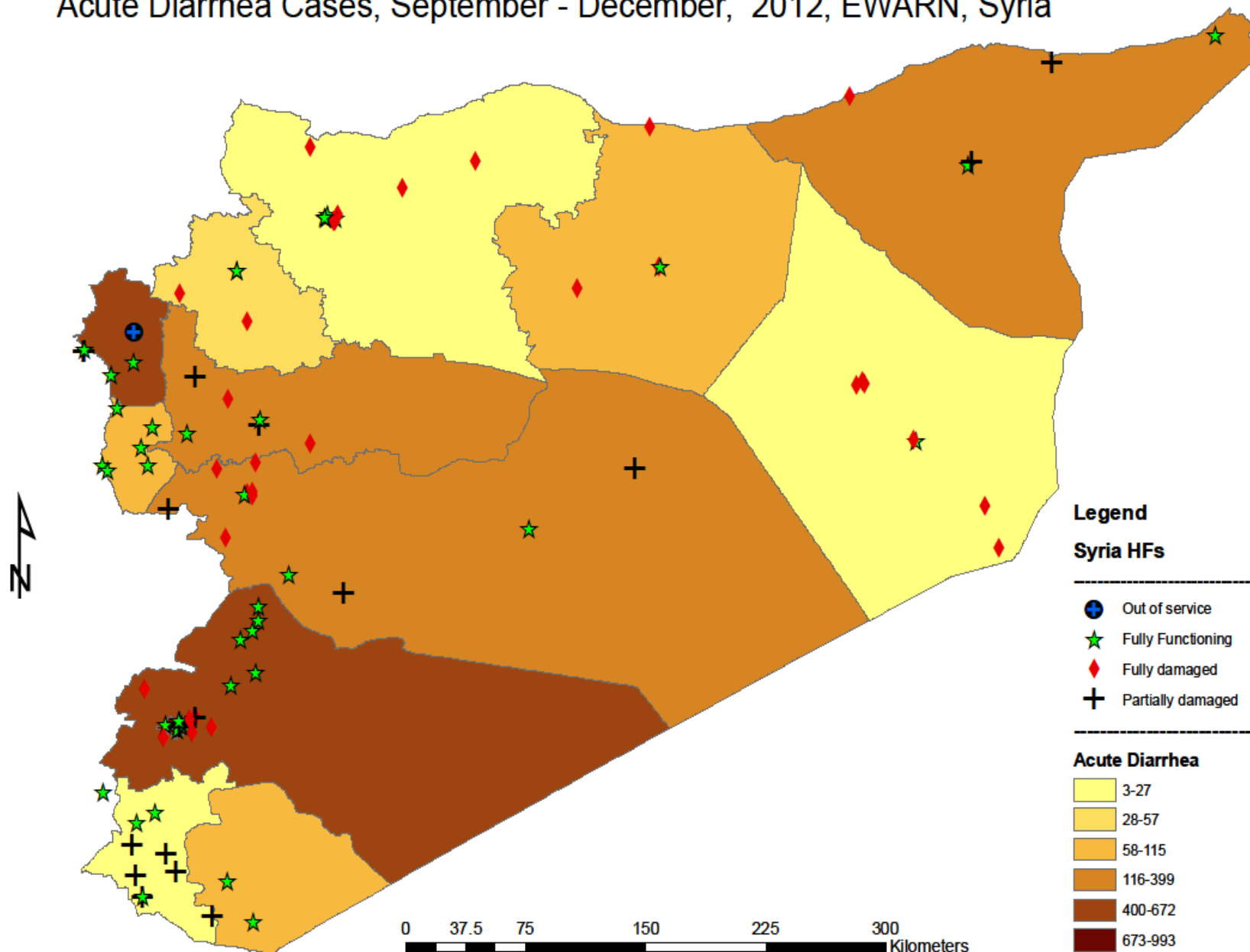
Data Source : WHO-Turkey
Date of Production: August, 2021
By: Negusu Worku : negusuworku@gmail.com



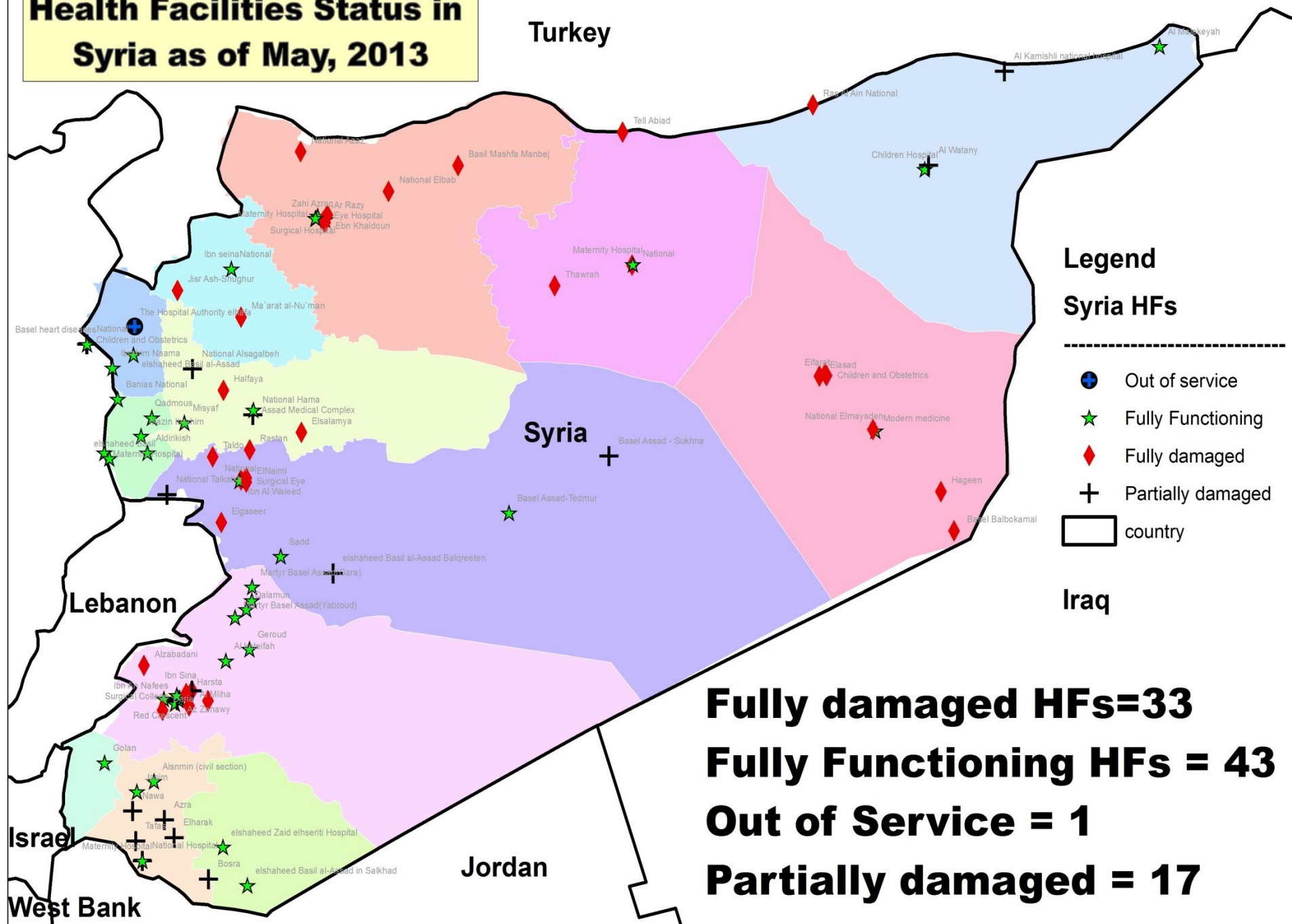
*From Kroman Chemical Industry **n=14** health facilities are available within 2kmR, of which **n=4 facilities** (1 health center and 3 pharmacies are available at CAYIROVA DISTRICT,

* From **3km radius from**, Kroman Chemical industry, **n=19** Health facilities are observed, of which **n=6 of them were from Chemetall Chemical industry** (n=3 Pharmacy and n=3 Health Center,

Acute Diarrhea Cases, September - December, 2012, EWARN, Syria



Health Facilities Status in Syria as of May, 2013



Bangladesh: Health Facilities by Type in Cox's Bazar's Camp Sites (As of 10 November 2018)



HEALTH SECTOR
COX'S BAZAR

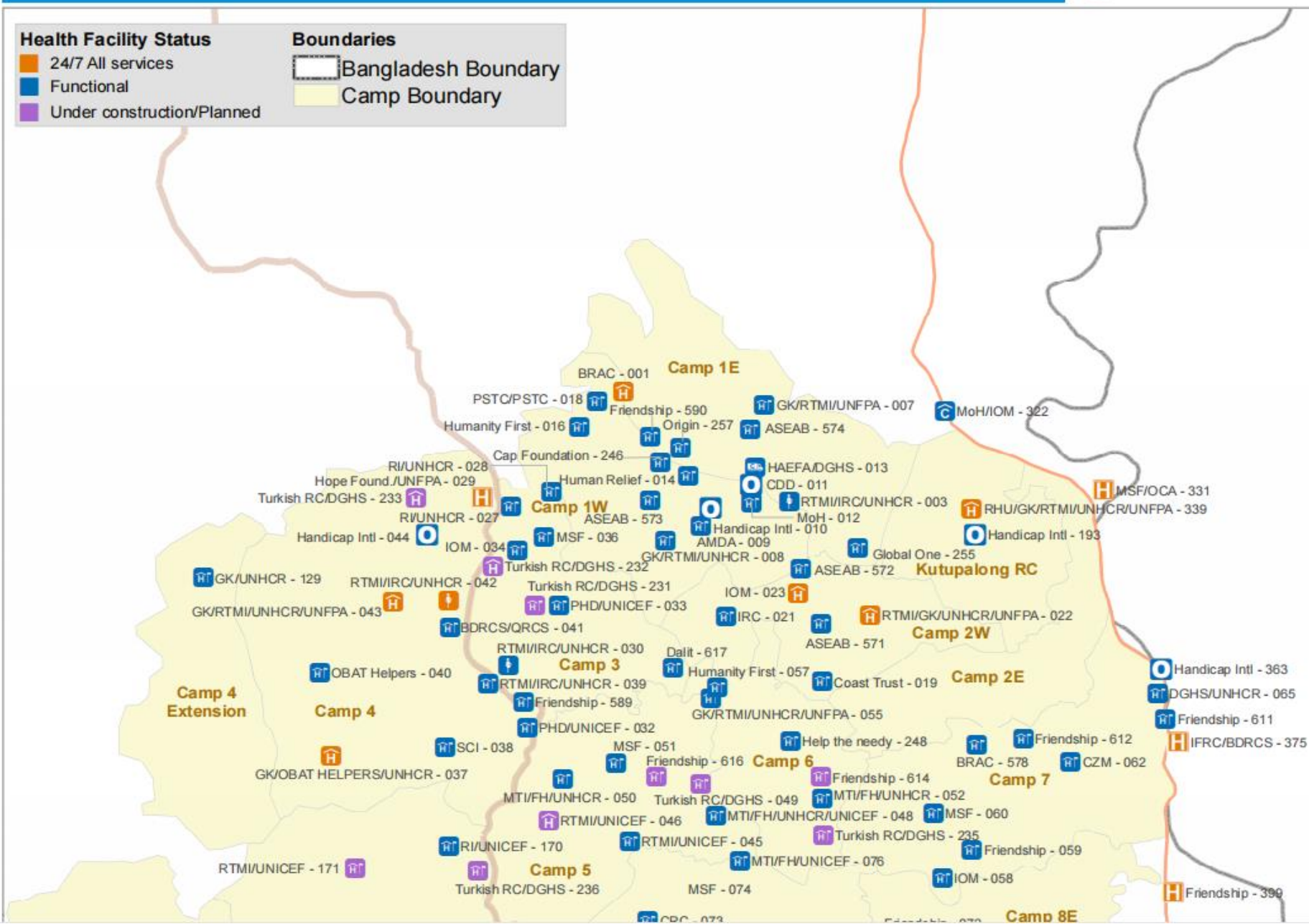


Health Facility Status

- 24/7 All services
- Functional
- Under construction/Planned

Boundaries

- Bangladesh Boundary
- Camp Boundary



References

1. Dr. Boris Pavlin, WHO-HQ, Switzerland, Geneva, Epidemiologist, pavlinb@who.int
2. Dr. Nsenga Ngoy nsengan@who.int Emergency Focal Person, WHO-AFRO, Congo, Brazzaville,
3. Dr. Sharmistha Mishra, MD, MSc, PhD, sharmistha.mishra@utoronto.ca
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4. Dr. Mohamed Diaaeldin Omar, UNICEF-Kenya, Nutrition and Health, mdomer@unicef.org
5. Dr. Anthony Omoko, Epidemiologist, tonyomoko@gmail.com
6. Dr. Mohamed ALI, ALIM@who.int , WHOHQ

Negusu Worku GEBRMICHAEL

Background

- IT Specialist
- International applied Epidemiology (Emory, USA)
- GIS
- VRAM
- M&E
- Data management
- Information management
- Data Scientist
- Web development